

Linear Attention: buying a memory that stops growing, and paying for it in recall

One-page concept summary · DataForge 2026, Pathway Track · companion to the interactive explainer *Where Memory Breaks*
For a data scientist meeting linear attention for the first time. Prerequisites: dot products, matrix–vector multiplication, and the standard query/key/value view of attention.

The design pressure

A Transformer scores each query against every earlier token: attention costs $O(T^2d)$ time and, at inference, forces the model to retain every key and value it has ever produced in a KV cache that costs $O(Td)$ and grows for as long as the conversation runs [1]. At a 100K-token context — no longer unusual in deployed systems — that cache alone can dwarf the model's own parameters in memory footprint. Linear attention removes the all-pairs comparison by reordering the computation: instead of comparing token pairs, it carries one fixed-size state that each token updates additively, $S \leftarrow S + \varphi(k_i) \otimes v_i$ and answers a query by reading $r(q) = \varphi(q)^T S$. Per-token cost becomes constant in sequence length, and the state has the same shape after ten tokens as after ten million. This is the mechanism, not merely a metaphor: our companion artifact runs both update rules on identical synthetic key/value streams so the trade-off below is something a reader watches happen, not just reads about.

Why this matters now

This is not a settled, textbook trade-off — it is being actively re-litigated in 2024–2026. Production systems now advertise context windows of hundreds of thousands to millions of tokens, making KV-cache memory a direct, dollar-denominated serving cost rather than a theoretical concern. That pressure is why four substantial papers in the last two years attack the same fixed-state failure mode from different angles: recall-throughput balancing [1], gated selective updates shipping in a production model [4], associative-memory reformulations of attention [3], and provably-bounded state norms [2]. A concept that mattered only in theory five years ago is now a line item in inference cost.

Four responses, and what each gives up

Approach	Memory rule	Recall behaviour	Evidence level
Softmax attention	Keeps every key–value pair; grows with context.	Exact — the reference mechanism. Cost, not accuracy, is the problem.	Ubiquitous production use.
State-space models (Mamba family)	Fixed-size recurrent state via a different, SSM-specific recurrence, not an attention rewrite.	Efficient, but reported to "struggle at recall" relative to attention-based models.	Author-reported benchmarks [1]; not the same mechanism family as BDH-GPU (see below).
Linear attention (kernel / normalized)	One fixed state, additive writes; feature map and denominator control scale.	Degrades once stored pairs exceed state capacity; normalization controls magnitude, not interference.	Author-reported benchmarks, incl. MQAR [1].
Gated DeltaNet	Fixed state; gating erases while the delta rule targets updates, so a write can edit rather than merely add.	Surpasses Mamba2 and DeltaNet on in-context retrieval, length extrapolation, long-context understanding.	Author benchmarks, ICLR 2025 [4]; separately reported in Qwen3-Next — a deployment signal, not an independent evaluation.
Variational linear attention	Fixed state; update recast as regularised online least squares with a Sherman–Morrison rank-1 penalty.	Cuts $\ S\ _F$ 109× at T=1,000; near-exact MQAR while pairs < head dimension; 62% at that boundary.	Proofs plus author-run experiments, 2026 preprint; no independent replication yet [2].
BDH-CQ contextual memory	Recurrent state updated by a <i>learned</i> function, once per demonstration rather than per token.	Adaptation happens in recurrent state, not weights — no parameter updates at inference.	Developer-reported.

Where BDH and BDH-CQ actually fit

Relevance is real but unequal. **BDH** (Dragon Hatchling) reformulates attention as synaptic memory: reasoning and memory share one computational fabric, associations are written by Hebbian updates between co-active neurons — "cells that fire together, wire together" made literal as a memory write — and activity is sparse and non-negative, roughly 5% of neurons active in reported runs, varying with input predictability rather than a fixed budget. Its GPU-oriented formulation, BDH-GPU, is built from ReLU-low-rank transformations *with* linear attention — the direct link to this concept — and is explicitly not a state-space model in the Mamba sense. This also touches an open problem the track brief names directly: consolidating useful *fast* state, the kind this toy accumulates, into durable *slow* parameters, versus treating it as disposable per-session memory.

BDH-CQ is the closer case. Its contextual memory is a general learned update $S_t \leftarrow U_\theta(S_{t-1}, D_t)$, and its report relates that memory to attention, fast-weight and linear-attention views of contextual association through an *additive special case* where state accumulates additively per demonstration. That special case — only that — is what our explainer reimplements, as a labelled synthetic toy on random vectors. We do not run BDH-CQ weights, do not claim U_θ is additive in general, and the artifact says so where a reader could otherwise conflate the two.

Continue here

- Arora, Eyuboglu, Zhang, et al. *Simple linear attention language models balance the recall-throughput tradeoff* (BASED). arXiv:2402.18668, 2024.
- Pandey & Singh. *Variational Linear Attention: Stable Associative Memory for Long-Context Transformers*. arXiv:2605.11196, 2026.
- Zhong, Xu, Ao & Shi (ByteDance Seed). *Understanding Transformer from the Perspective of Associative Memory*. arXiv:2505.19488, 2025.
- Yang, Kautz & Hatamizadeh. *Gated Delta Networks: Improving Mamba2 with Delta Rule*. arXiv:2412.06464, ICLR 2025.
- Arora et al. *Zoology: Measuring and Improving Recall in Efficient Language Models*. arXiv:2312.04927, ICLR 2024 — origin of the MQAR task our explainer runs live.
- Pathway. *The Dragon Hatchling* and the *BDH-CQ* technical report — primary sources for the BDH claims above.

What the change costs

The state is a superposition: every association is summed into the same matrix, so nothing is stored separately. Probing it with the key that wrote a fact returns that fact *plus* a weighted sum of every other stored value:

$$r(k_j) = (k_j^T k_i) v_i + \sum_{j \neq i} (k_j^T k_i) v_j$$

The central claim our explainer teaches: a fixed-shape state can absorb an unbounded stream of key–value facts, but because all writes are superposed, retrieval degrades as facts accumulate or as keys grow more similar — and enlarging the state postpones that failure rather than removing it.

The geometry is simple: a key's dot product with itself grows with dimension d , while its dot product with an unrelated key grows only with $\sqrt{d}$ for random vectors — the "self-signal" dominates when facts are few and well-separated, but the sum of many small cross-terms eventually rivals it. Arora et al. identify this as a direct state-size/recall trade-off and show fixed-state alternatives such as H3, Mamba and RWKV "struggle at recall"; their BASED architecture makes state size a dial traversing the recall–memory Pareto frontier, beating comparable sub-quadratic models by 10.36 accuracy points on recall-intensive tasks [1]. Pandey & Singh sharpen the cause: the state grows as $O(T)$ in Frobenius norm, "causing progressive interference between stored associations" [2]. Zhong et al. formalise the same separation as a retrieval signal-to-noise ratio under an associative-memory reading of attention [3]. This is the specific, falsifiable mechanism a reader should demand before accepting any "fixed memory, same quality" claim.

The most important open question

Not whether gating or regularisation helps — both demonstrably do — but whether *any* fixed-size state can hold what a long-context task requires without growing the state or smuggling exact storage back in through a side channel (a sliding window, a retrieval index, a hybrid layer). Both leading fixes concede the capacity boundary rather than dissolve it: VLA's accuracy holds while stored pairs stay below the per-head dimension and drops to 62% at that boundary [2], and BASED's own answer is explicitly a hybrid — linear *plus* sliding-window attention, state size left as a tunable knob rather than a solved constant [1]. Independent reproduction is the missing evidence throughout: the 2026 variational and BDH/BDH-CQ results are, as of writing, reported by their own authors, not yet replicated externally.

How the artifact makes this checkable

The companion explainer runs both retrieval mechanisms live, on identical synthetic key/value streams (10-value vocabulary, random Gaussian keys, no learned projections), and lets the reader move three real variables — fact count, state size, key similarity — while accuracy, a live Frobenius-norm readout, and a per-query signal-vs-cross-talk breakdown update within one frame. "Reshuffle" redraws a fresh seed so a sceptical reader can confirm the numbers are computed, not replayed. The claim above is falsifiable in under a minute: shrink the state, grow the fact count, and watch linear retrieval fall away from softmax's while the state's norm climbs — or fail to, which would falsify it.